

# 13 to 17

## Set 13. Dataset: Geographic Data

**13.1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.**

```
install.packages("ggplot2")
```

```
install.packages("maps")
```

```
library(ggplot2)
```

```
library(maps)
```

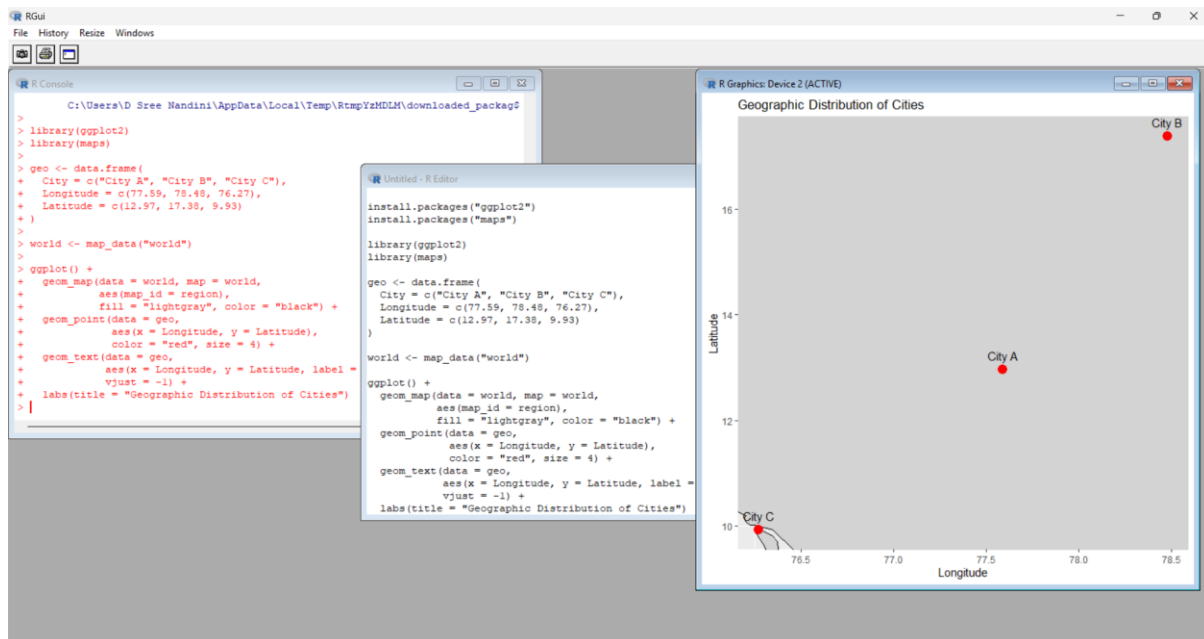
```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Longitude = c(77.59, 78.48, 76.27),  
  Latitude = c(12.97, 17.38, 9.93)  
)
```

```
world <- map_data("world")
```

```
ggplot() +  
  geom_map(data = world, map = world,  
    aes(map_id = region),  
    fill = "lightgray", color = "black") +  
  geom_point(data = geo,  
    aes(x = Longitude, y = Latitude),  
    color = "red", size = 4) +  
  geom_text(data = geo,  
    aes(x = Longitude, y = Latitude, label = City),
```

vjust = -1) +

labs(title = "Geographic Distribution of Cities")

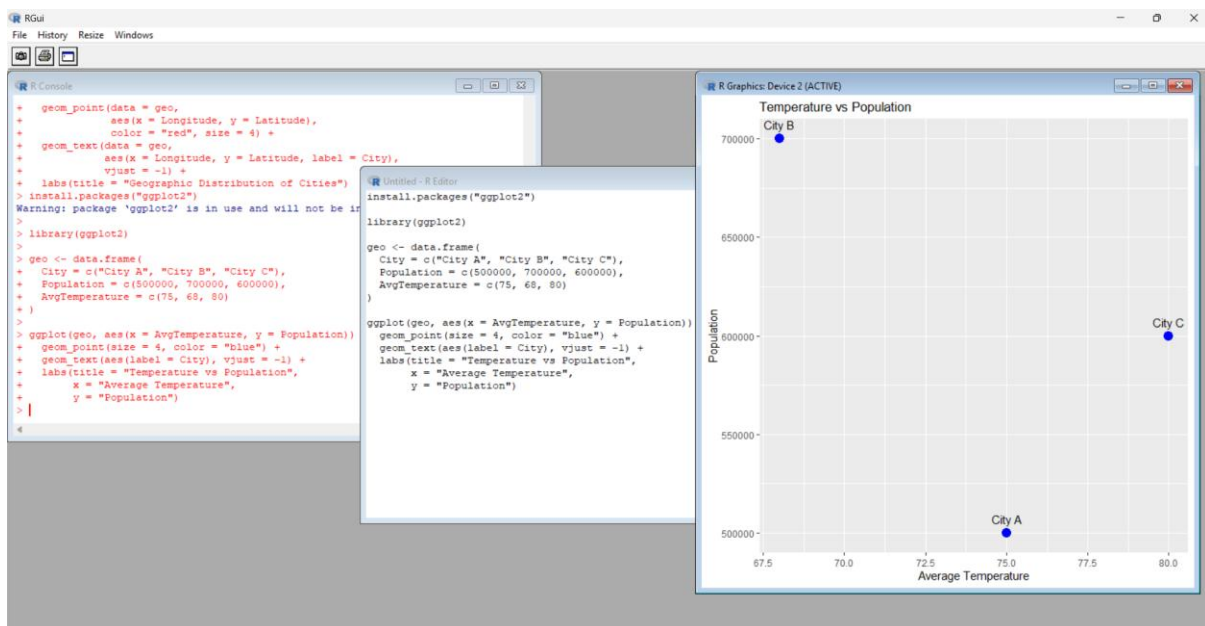


### 13.2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```
install.packages("ggplot2")
```

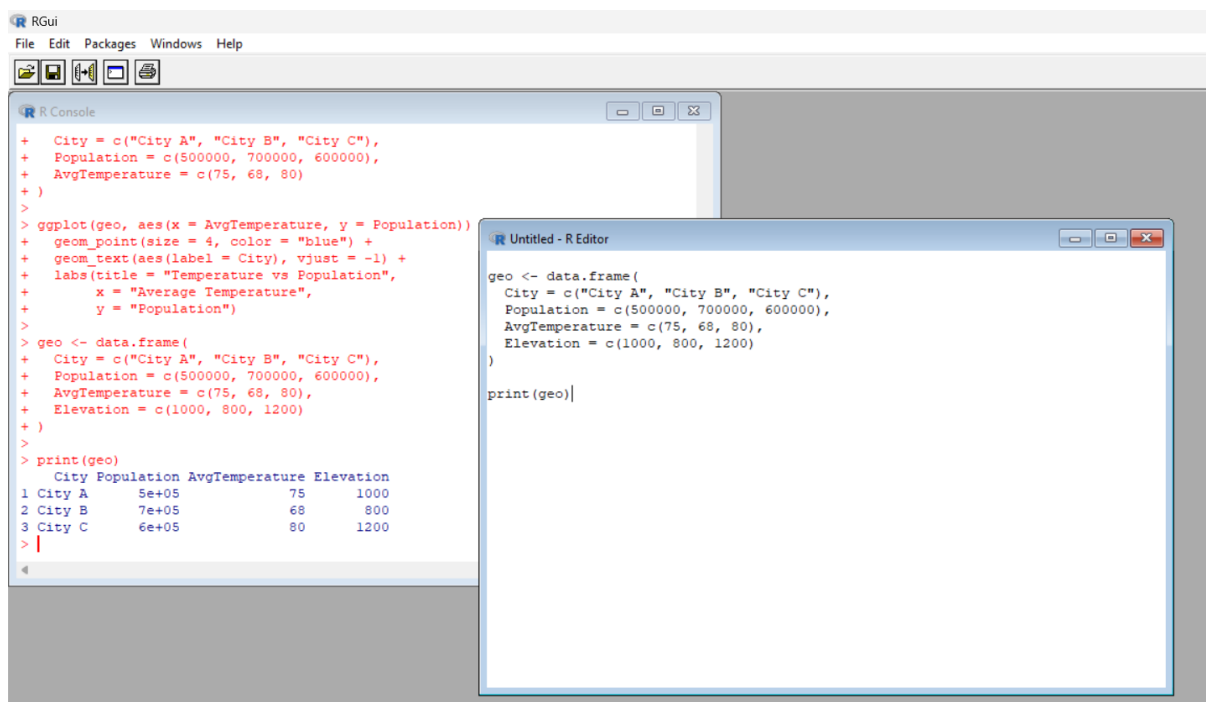
```
library(ggplot2)
```

```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  AvgTemperature = c(75, 68, 80)  
)  
  
ggplot(geo, aes(x = AvgTemperature, y = Population)) +  
  geom_point(size = 4, color = "blue") +  
  geom_text(aes(label = City), vjust = -1) +  
  labs(title = "Temperature vs Population",  
        x = "Average Temperature",  
        y = "Population")
```



### 13.3. Build a table to display the geographic data for each city. Label the table elements.

```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  AvgTemperature = c(75, 68, 80),  
  Elevation = c(1000, 800, 1200)  
)  
  
print(geo)
```



**13. 4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.**

```
install.packages("shiny")
```

```
install.packages("leaflet")
```

```
install.packages("ggplot2")
```

```
install.packages("DT")
```

```
library(shiny)
```

```
library(leaflet)
```

```
library(ggplot2)
```

```
library(DT)
```

```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  AvgTemperature = c(75, 68, 80),  
  Elevation = c(1000, 800, 1200),  
  Longitude = c(77.59, 78.48, 76.27),  
  Latitude = c(12.97, 17.38, 9.93)  
)
```

```
ui <- fluidPage(  
  titlePanel("Geographic Data Dashboard"),  
  
  fluidRow(  
    column(6,  
      leafletOutput("map", height = 400)  
    ),  
  ),  
)
```

```

column(6,
  plotOutput("scatter", height = 400)
)
),

```

```

fluidRow(
  column(12,
    dataTableOutput("table")
  )
)
)
)

```

```

server <- function(input, output) {

```

```

  output$map <- renderLeaflet({
    leaflet(geo) %>%
      addTiles() %>%
      addMarkers(
        lng = ~Longitude,
        lat = ~Latitude,
        popup = ~City
      )
  })

```

```

  output$scatter <- renderPlot({
    ggplot(geo, aes(x = AvgTemperature, y = Population, label = City)) +
      geom_point(color = "blue", size = 4) +
      geom_text(vjust = -1) +
      labs(

```

```

    title = "Temperature vs Population",
    x = "Average Temperature",
    y = "Population"
  )
})

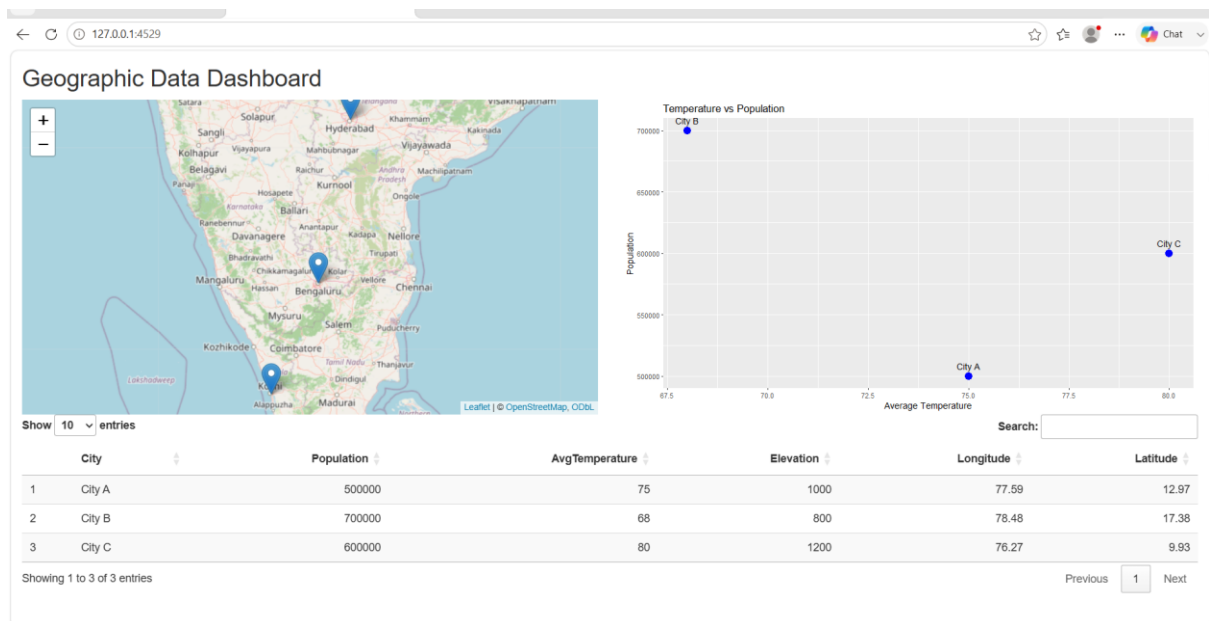
```

```

output$table <- renderDataTable({
  geo
})
}

```

```
shinyApp(ui = ui, server = server)
```



## Set 14. Dataset: Survey Results

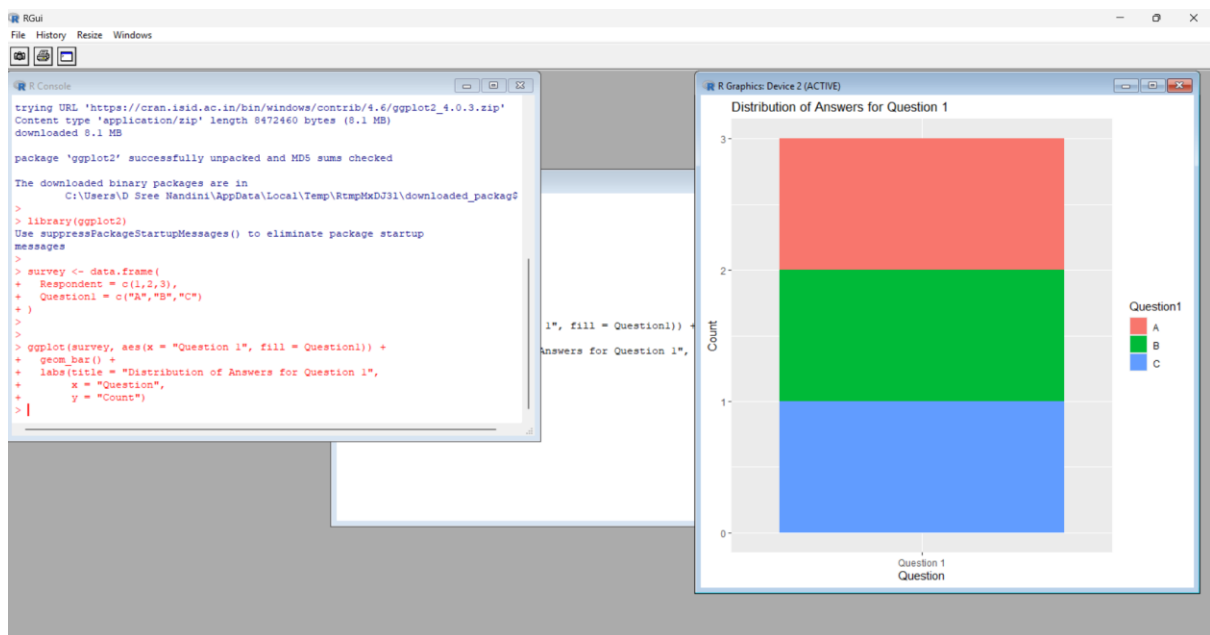
### 14.1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
survey <- data.frame(  
  Respondent = c(1,2,3),  
  Question1 = c("A","B","C")  
)
```

```
ggplot(survey, aes(x = "Question 1", fill = Question1)) +  
  geom_bar() +  
  labs(title = "Distribution of Answers for Question 1",  
        x = "Question",  
        y = "Count")
```



## 14.2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
install.packages("fmsb")
```

```
library(fmsb)
```

```
data <- data.frame(
```

```
  Q1 = c(3,1,2),
```

```
  Q2 = c(2,3,1),
```

```
  Q3 = c(1,2,3)
```

```
)
```

```
radar <- rbind(
```

```
  c(3,3,3),
```

```
  c(1,1,1),
```

```
  data
```

```
)
```

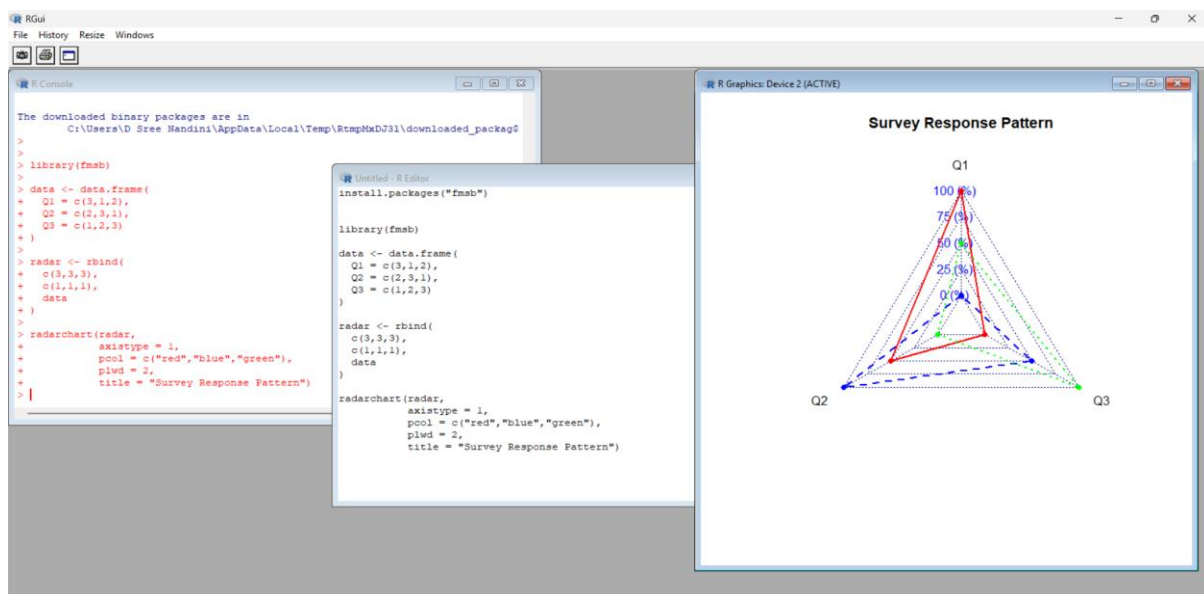
```
radarchart(radar,
```

```
  axistype = 1,
```

```
  pcol = c("red","blue","green"),
```

```
  plwd = 2,
```

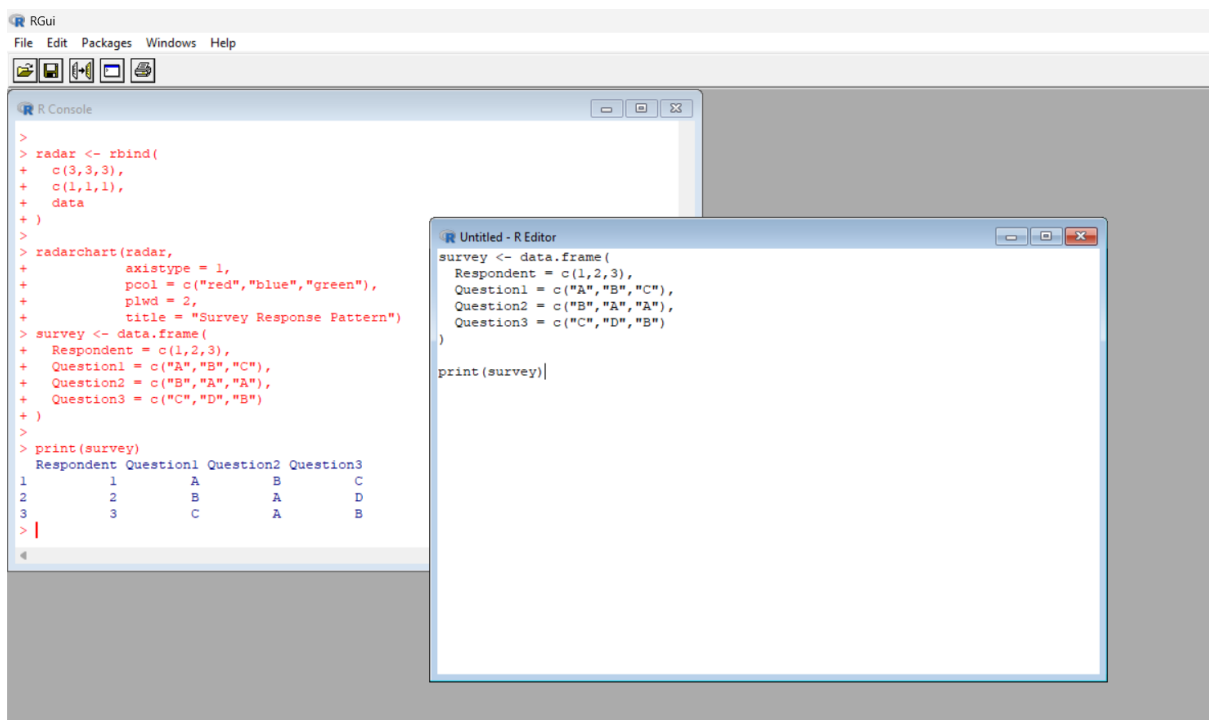
```
  title = "Survey Response Pattern")
```



### 14.3. Build a table to show the survey response data for each respondent. Label the table elements.

```
survey <- data.frame(  
  Respondent = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
print(survey)
```



**14.4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.**

```
install.packages("shiny")
```

```
install.packages("ggplot2")
```

```
install.packages("DT")
```

```
library(shiny)
```

```
library(ggplot2)
```

```
library(DT)
```

```
survey <- data.frame(  
  Respondent = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
ui <- fluidPage(  
  titlePanel("Survey Results Dashboard"),
```

```
  fluidRow(  
    column(6,  
      plotOutput("barplot")),  
    column(6,  
      tableOutput("table"))  
  )  
)
```

```
server <- function(input, output){
```

```

output$barplot <- renderPlot({
  ggplot(survey, aes(x="Question 1", fill=Question1)) +
    geom_bar() +
    labs(title="Stacked Bar Chart")
})

```

```

output$table <- renderTable({
  survey
})

```

```

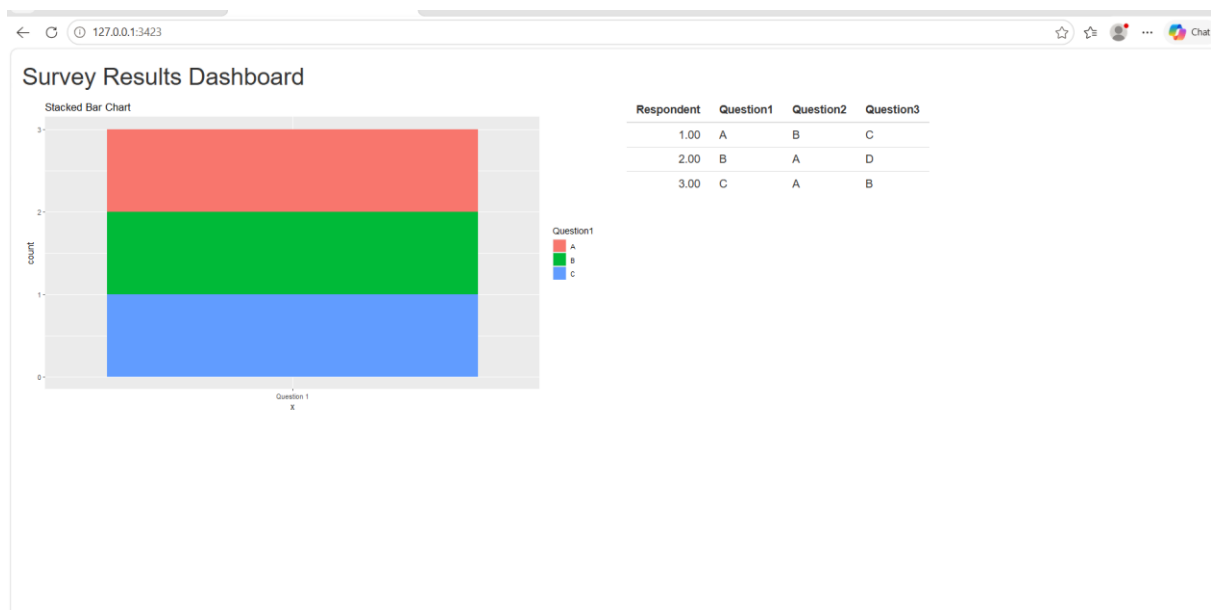
}

```

```

shinyApp(ui, server)

```



## Set 15: Product Sales Analysis

**15.1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.**

```
library(ggplot2)

library(tidyr)

sales <- data.frame(

  Product = c("Product A", "Product B", "Product C"),

  January = c(2000, 1500, 1200),

  February = c(2200, 1800, 1400),

  March = c(2400, 1600, 1100)

)
```

```
sales_long <- pivot_longer(

  sales,

  cols = January:March,

  names_to = "Month",

  values_to = "Sales"

)
```

```
ggplot(sales_long,

  aes(x = Product,

    y = Sales,

    fill = Month)) +

  geom_bar(stat = "identity",

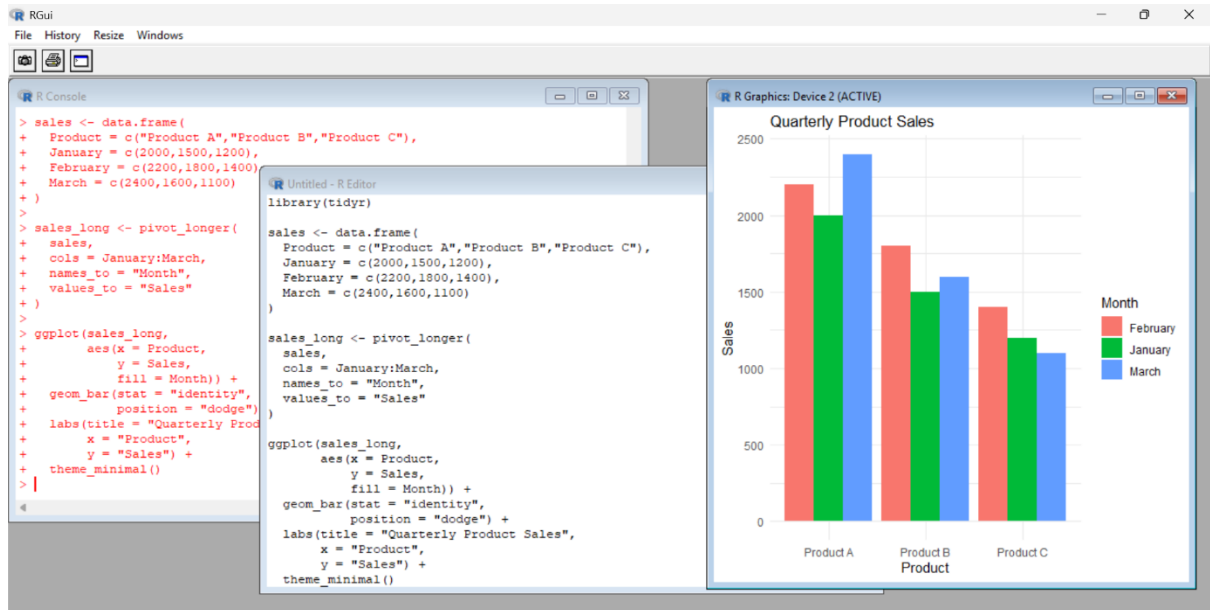
    position = "dodge") +

  labs(title = "Quarterly Product Sales",

    x = "Product",

    y = "Sales") +

  theme_minimal()
```



## 15.2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
library(ggplot2)

library(tidyr)

sales_long <- pivot_longer(

  sales,

  cols = January:March,

  names_to = "Month",

  values_to = "Sales"

)

ggplot(sales_long,

  aes(x = Month,

      y = Sales,

      fill = Product,

      group = Product)) +

  geom_area(alpha = 0.7) +

  labs(title = "Overall Sales Trend",

      x = "Month",

      y = "Sales") +

  theme_minimal()
```



### 15.3. Build a table to show the monthly sales data for each product. Label the table elements.

```
sales <- data.frame(  
  Product = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
print(sales)
```

The screenshot shows an R environment with two windows. The 'R Console' window on the left displays the execution of R code. The code first creates a ggplot titled 'Overall Sales Trend' with 'Month' on the x-axis and 'Sales' on the y-axis, faceted by 'Product'. It then creates a data frame 'sales' with columns 'Product', 'January', 'February', and 'March'. Finally, it prints the 'sales' data frame, resulting in a table with 3 rows and 4 columns.

```
R Console  
+ )  
>  
> ggplot(sales_long,  
+   aes(x = Month,  
+     y = Sales,  
+     fill = Product,  
+     group = Product)) +  
+   geom_area(alpha = 0.7) +  
+   labs(title = "Overall Sales Trend",  
+     x = "Month",  
+     y = "Sales") +  
+   theme_minimal()  
> sales <- data.frame(  
+   Product = c("Product A","Product B","Product C"),  
+   January = c(2000,1500,1200),  
+   February = c(2200,1800,1400),  
+   March = c(2400,1600,1100)  
+ )  
>  
> print(sales)  
  Product January February March  
1 Product A   2000    2200  2400  
2 Product B   1500    1800  1600  
3 Product C   1200    1400  1100  
> |
```

The 'Untitled - R Editor' window on the right shows the source code for the 'sales' data frame:

```
Untitled - R Editor  
sales <- data.frame(  
  Product = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
print(sales)|
```

**15.4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.**

```
library(ggplot2)
```

```
library(gridExtra)
```

```
library(tidyr)
```

```
sales <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
sales_long <- pivot_longer(  
  sales,  
  cols = January:March,  
  names_to = "Month",  
  values_to = "Sales"  
)
```

```
# Grouped Bar Chart
```

```
p1 <- ggplot(sales_long,  
  aes(Product, Sales, fill = Month)) +  
  geom_bar(stat = "identity",  
    position = "dodge") +  
  labs(title = "Grouped Bar Chart") +  
  theme_minimal()
```

```
# Stacked Area Chart
```

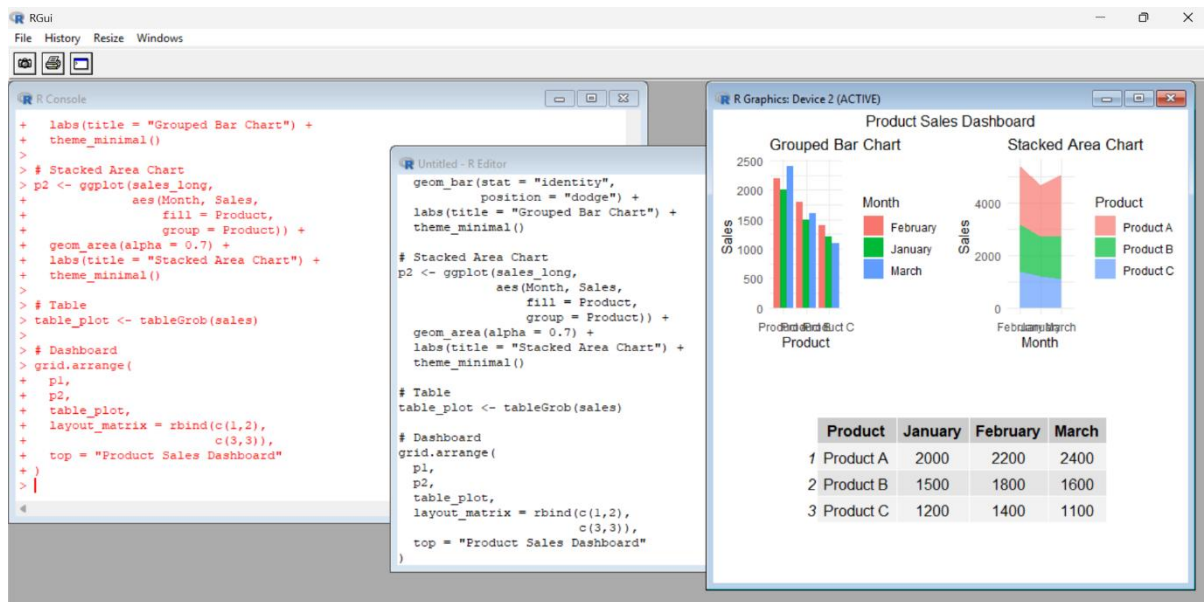
```
p2 <- ggplot(sales_long,
             aes(Month, Sales,
                 fill = Product,
                 group = Product)) +
  geom_area(alpha = 0.7) +
  labs(title = "Stacked Area Chart") +
  theme_minimal()
```

```
# Table
```

```
table_plot <- tableGrob(sales)
```

```
# Dashboard
```

```
grid.arrange(
  p1,
  p2,
  table_plot,
  layout_matrix = rbind(c(1,2),
                        c(3,3)),
  top = "Product Sales Dashboard"
)
```



## Set 16 – Customer Demographics Analysis

**16.1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.**

```
library(ggplot2)
```

```
customers <- data.frame(
```

```
  Customer = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```
  Gender = c("Female","Male","Female"),
```

```
  Income = c(50000,60000,75000)
```

```
)
```

```
ggplot(customers,
```

```
  aes(x=factor(Customer),
```

```
    y=Age,
```

```
    fill=factor(Customer))) +
```

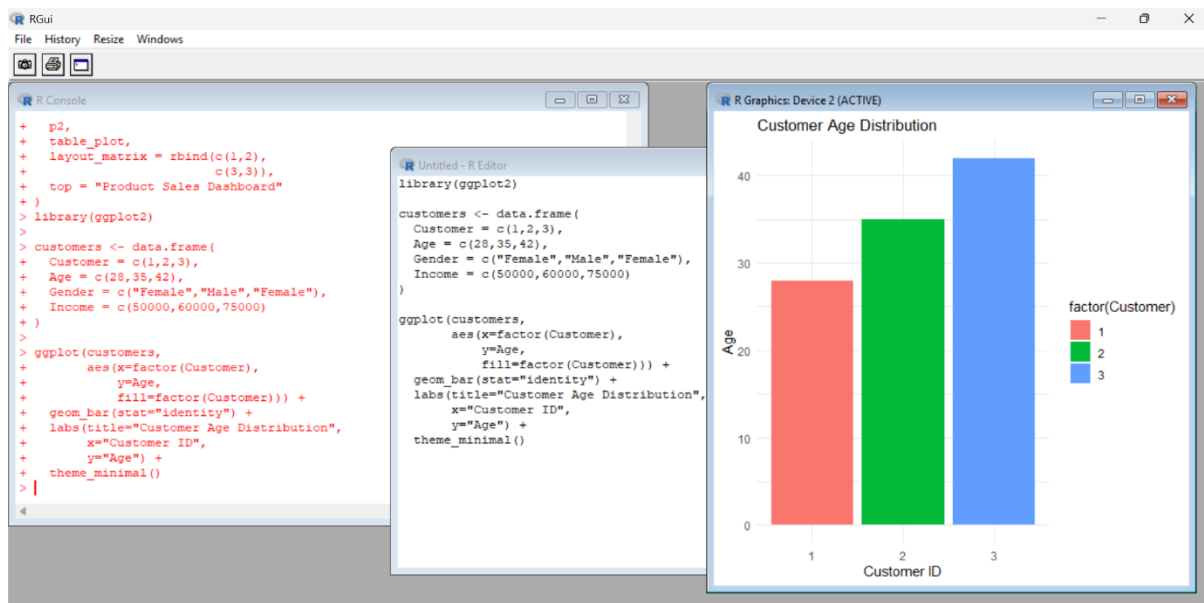
```
geom_bar(stat="identity") +
```

```
labs(title="Customer Age Distribution",
```

```
  x="Customer ID",
```

```
  y="Age") +
```

theme\_minimal()



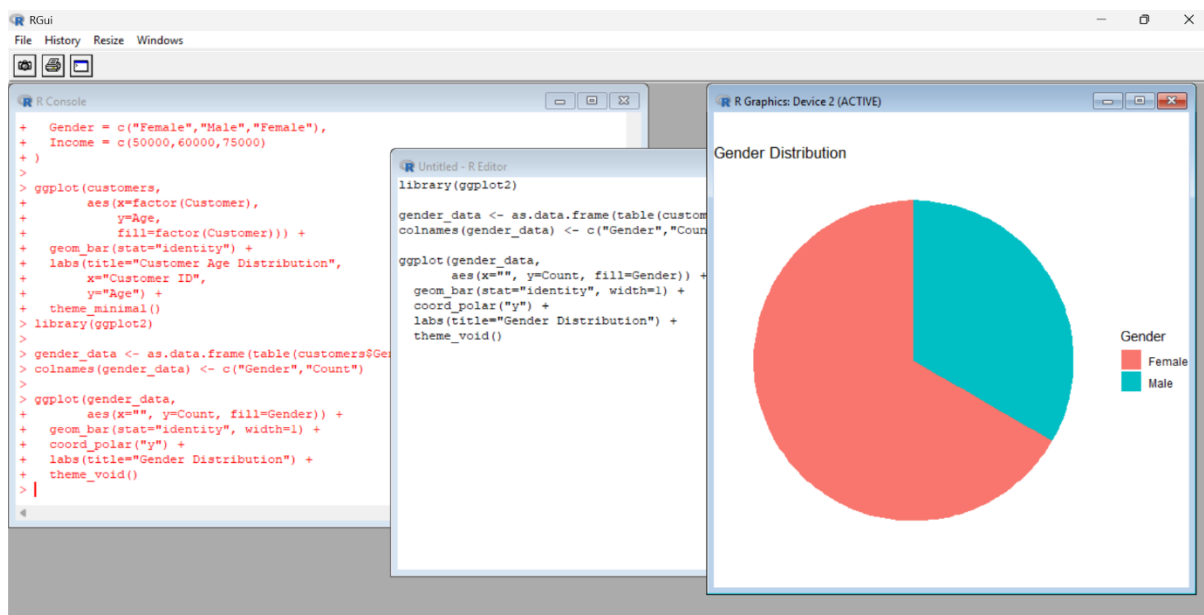
## 16.2. Generate a pie chart to represent the distribution of customers by gender.

```
library(ggplot2)

gender_data <- as.data.frame(table(customers$Gender))

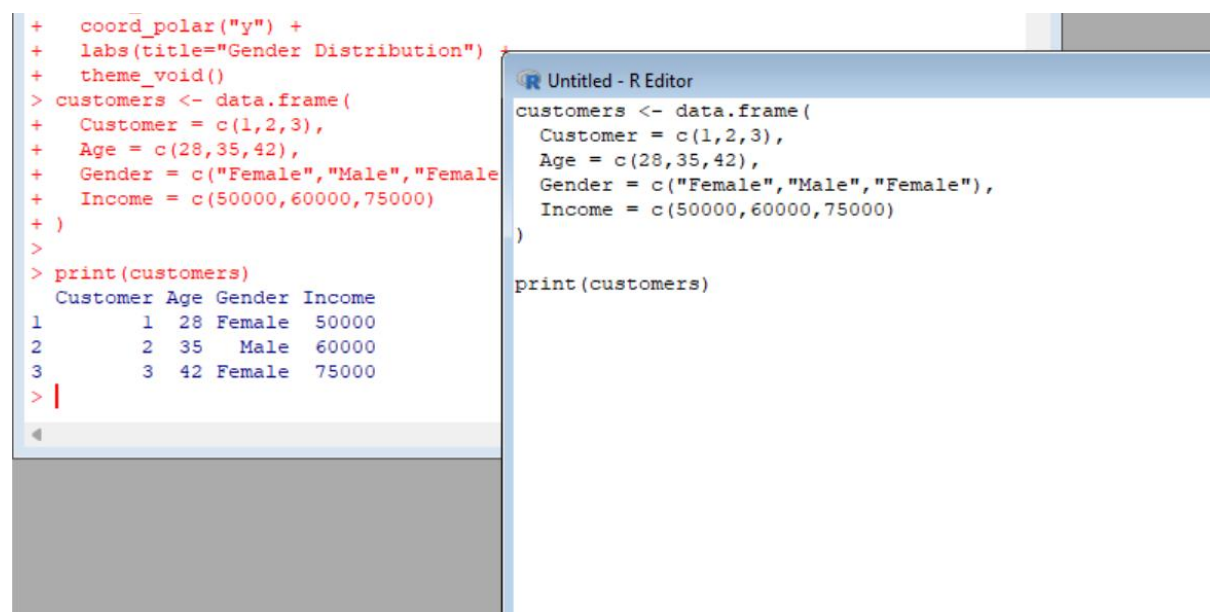
colnames(gender_data) <- c("Gender", "Count")

ggplot(gender_data,
       aes(x="", y=Count, fill=Gender)) +
  geom_bar(stat="identity", width=1) +
  coord_polar("y") +
  labs(title="Gender Distribution") +
  theme_void()
```



**16.3. Build a table to show the demographic information for each customer. Label the table elements.**

```
customers <- data.frame(  
  Customer = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
print(customers)
```



The image shows a screenshot of an R environment. On the left is the R console, and on the right is the R Editor window titled 'Untitled - R Editor'.

**R Console:**

```
+ coord_polar("y") +  
+ labs(title="Gender Distribution") +  
+ theme_void()  
> customers <- data.frame(  
+   Customer = c(1,2,3),  
+   Age = c(28,35,42),  
+   Gender = c("Female","Male","Female"),  
+   Income = c(50000,60000,75000)  
+ )  
>  
> print(customers)  
  Customer Age Gender Income  
1         1  28 Female 50000  
2         2  35   Male 60000  
3         3  42 Female 75000  
> |
```

**R Editor:**

```
customers <- data.frame(  
  Customer = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
print(customers)
```

**16.4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.**

```
library(ggplot2)

library(gridExtra)

customers <- data.frame(

  Customer = c(1,2,3),

  Age = c(28,35,42),

  Gender = c("Female","Male","Female"),

  Income = c(50000,60000,75000)

)

p1 <- ggplot(customers,

  aes(x=factor(Customer),

    y=Age,

    fill=factor(Customer))) +

  geom_bar(stat="identity") +

  labs(title="Customer Age Distribution",

    x="Customer ID",

    y="Age") +

  theme_minimal()

gender_data <- as.data.frame(table(customers$Gender))

colnames(gender_data) <- c("Gender","Count")

p2 <- ggplot(gender_data,

  aes(x="", y=Count, fill=Gender)) +

  geom_bar(stat="identity", width=1) +

  coord_polar("y") +

  labs(title="Gender Distribution") +

  theme_void()

table_plot <- tableGrob(customers)

grid.arrange(
```

p1,

p2,

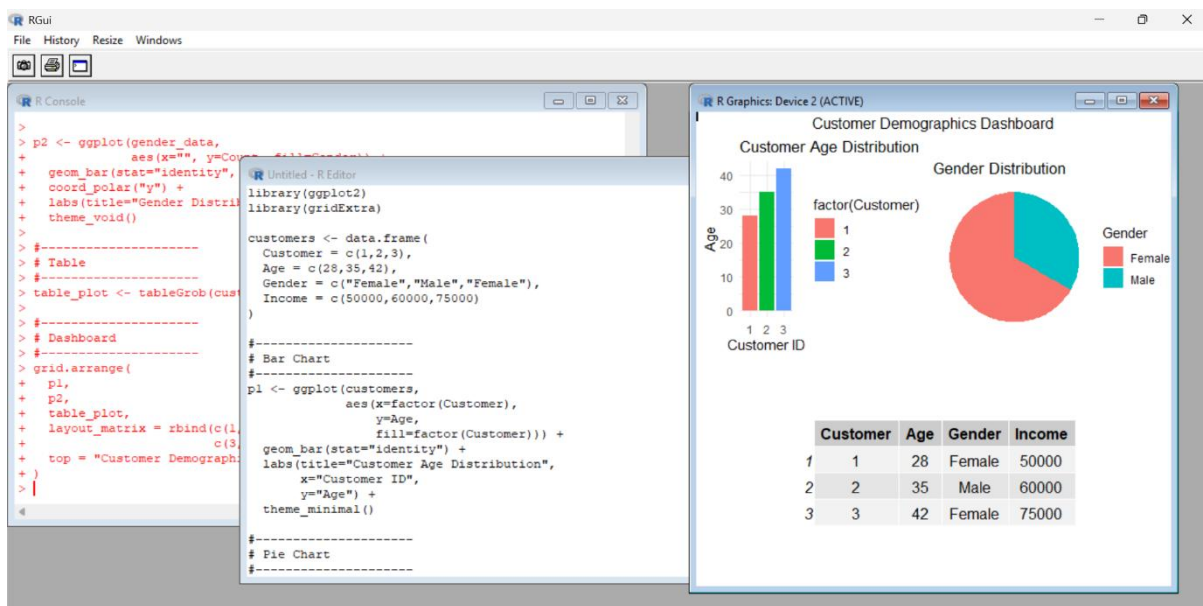
table\_plot,

layout\_matrix = rbind(c(1,2),

c(3,3)),

top = "Customer Demographics Dashboard"

)



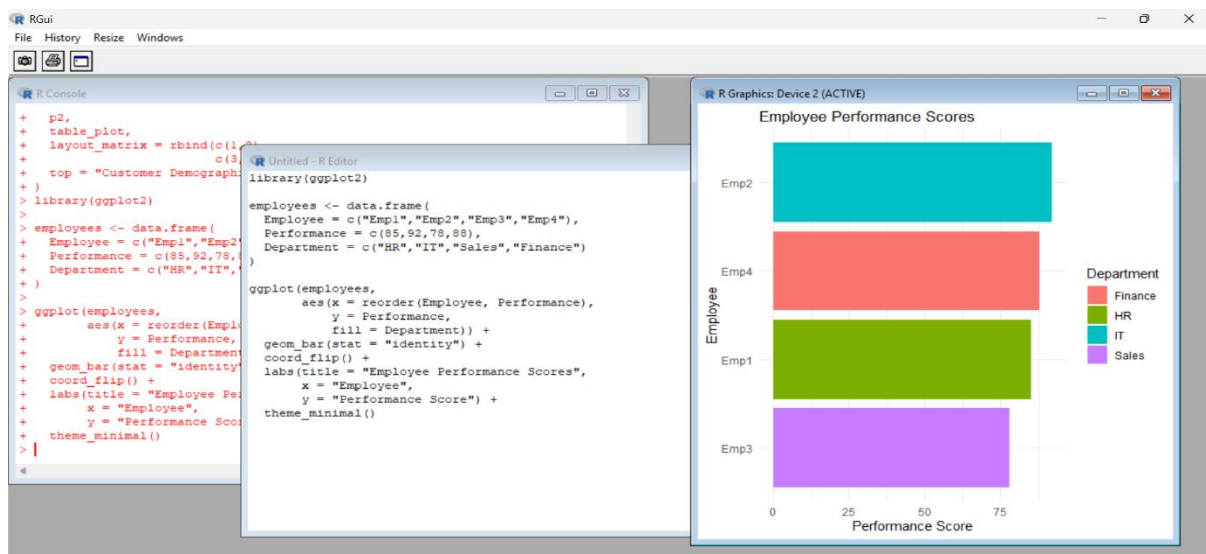
## Set 17 – Employee Performance Analysis

**17.1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.**

```
library(ggplot2)
```

```
employees <- data.frame(  
  Employee = c("Emp1","Emp2","Emp3","Emp4"),  
  Performance = c(85,92,78,88),  
  Department = c("HR","IT","Sales","Finance")  
)
```

```
ggplot(employees,  
  aes(x = reorder(Employee, Performance),  
    y = Performance,  
    fill = Department)) +  
  geom_bar(stat = "identity") +  
  coord_flip() +  
  labs(title = "Employee Performance Scores",  
    x = "Employee",  
    y = "Performance Score") +  
  theme_minimal()
```



## 17.2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
library(ggplot2)
```

```
employees$Experience <- c(2,5,3,4)
```

```
ggplot(employees,
```

```
  aes(x = Experience,
```

```
      y = Performance,
```

```
      size = Performance,
```

```
      color = Department)) +
```

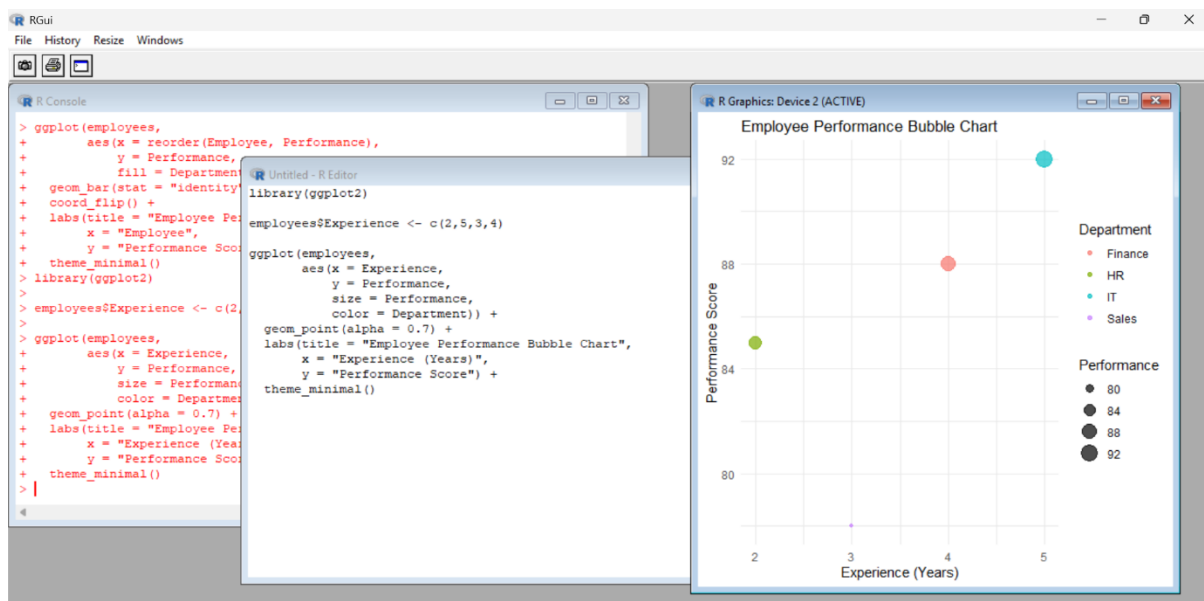
```
geom_point(alpha = 0.7) +
```

```
labs(title = "Employee Performance Bubble Chart",
```

```
      x = "Experience (Years)",
```

```
      y = "Performance Score") +
```

```
theme_minimal()
```



### 17.3. Build a table to display the performance data for each employee. Label the table elements.

```
employees <- data.frame(  
  Employee = c("Emp1","Emp2","Emp3","Emp4"),  
  Performance = c(85,92,78,88),  
  Department = c("HR","IT","Sales","Finance"),  
  Experience = c(2,5,3,4)  
)  
print(employees)
```

```
+ labs(title = "Employee Performance Bubble Chart",  
+       x = "Experience (Years)",  
+       y = "Performance Score") +  
+ theme_minimal()  
> employees <- data.frame(  
+   Employee = c("Emp1","Emp2","Emp3","Emp4"),  
+   Performance = c(85,92,78,88),  
+   Department = c("HR","IT","Sales","Finance"),  
+   Experience = c(2,5,3,4)  
+ )  
>  
> print(employees)  
  Employee Performance Department Experience  
1    Emp1         85         HR          2  
2    Emp2         92         IT          5  
3    Emp3         78        Sales          3  
4    Emp4         88        Finance         4  
> |
```

```
Untitled - R Editor  
employees <- data.frame(  
  Employee = c("Emp1","Emp2","Emp3","Emp4"),  
  Performance = c(85,92,78,88),  
  Department = c("HR","IT","Sales","Finance"),  
  Experience = c(2,5,3,4)  
)  
print(employees)|
```

**17.4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.**

```
library(ggplot2)

library(gridExtra)

employees <- data.frame(

  Employee = c("Emp1","Emp2","Emp3","Emp4"),
  Performance = c(85,92,78,88),
  Department = c("HR","IT","Sales","Finance"),
  Experience = c(2,5,3,4)
)

p1 <- ggplot(employees,
  aes(x = reorder(Employee, Performance),
    y = Performance,
    fill = Department)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(title = "Employee Performance",
    x = "Employee",
    y = "Score") +
  theme_minimal()

p2 <- ggplot(employees,
  aes(x = Experience,
    y = Performance,
    size = Performance,
    color = Department)) +
  geom_point(alpha = 0.8) +
  labs(title = "Bubble Chart") +
  theme_minimal()

table_plot <- tableGrob(employees)
```

```

grid.arrange(
  p1,
  p2,
  table_plot,
  layout_matrix = rbind(c(1,2),
                          c(3,3)),
  top = "Employee Performance Dashboard"
)

```

